

# Clébio Júnior

## DATA SCIENTIST

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## PROFESSIONAL SUMMARY

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Data Scientist with over 5 years of experience applying machine learning, statistics and data analysis to solve real business problems with measurable impact. Proven track record in predictive modeling for credit scoring, sales forecasting, logistics demand forecasting, anomaly detection, customer segmentation and consumer behavior analysis across the credit risk, legal and retail sectors. Strong experience with both structured and unstructured data, including text extraction and processing with NLP and OCR. Master's degree in Natural Sciences, with a solid quantitative background in statistics, modeling and translating results into business decisions.

## TECHNICAL SKILLS

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- **Programming Languages:** Python, SQL, C/C++
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, Predictive Modeling, Feature Engineering, Feature Selection, Cross-Validation, Hyperparameter Optimization
- **Algorithms:** Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, KNN, SVM, K-Means, Gaussian Mixture Models
- **Statistics and Analysis:** Descriptive and Inferential Statistics, Exploratory Data Analysis (EDA), Time Series Analysis, Hypothesis Testing, Outlier Detection, Cohort Analysis
- **NLP and Unstructured Data:** spaCy, NLTK, Embeddings, Text Classification, Tokenization, Lemmatization, OCR (Tesseract), pdfplumber, Regular Expressions (Regex)
- **Libraries and Tools:** pandas, NumPy, scikit-learn, SciPy, Matplotlib, Seaborn, Plotly, Optuna, SHAP, Boruta, Jupyter Notebook
- **Engineering and Deployment:** Flask, REST APIs, Docker, Git, GitHub, Linux
- **Methodologies:** CRISP-DM, Business Metrics, A/B Testing, Agile Methodologies

## PROFESSIONAL EXPERIENCE

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### Data Scientist — Vert Analytics

October/2024 – Present | Remote

- Lead data exploration, analysis and modeling projects on structured and unstructured data to generate strategic insights, delivering results through interactive SAS dashboards.
- Developed an NLP-based recommendation system for legal defenses using embeddings and FAISS vector search, reducing manual research time and increasing the accuracy of the team's decisions.
- Built automated pipelines to extract and clean data from PDF files, including scanned documents, using pdfplumber, Tesseract OCR and regular expressions, structuring information that was previously unavailable for analysis.
- Implemented a text analysis solution with spaCy to classify customer complaints on social media, applying tokenization, lemmatization, stopword removal and grammar correction to identify recurring themes.
- Perform time series analysis to identify patterns and anomalies, supporting product, customer service and operations decisions.

### Data Scientist — Datarisk

January/2022 – August/2024 | Remote

- Developed a machine learning credit scoring model to estimate repayment probability from customers' financial history, improving credit approval criteria and the efficiency of risk management.
- Built a regression model to forecast sales volume from historical data, with preprocessing, feature engineering and hyperparameter optimization, which went on to support strategic planning for the commercial team.
- Applied clustering algorithms for customer base segmentation, identifying distinct behavioral profiles and enabling targeted relationship strategies.
- Executed the full analytical cycle of data cleaning, exploration and analysis, translating business problems into machine learning solutions and presenting results to non-technical stakeholders.

### Data Scientist — Be.X!

March/2021 – January/2022 | Remote

- Developed analytics projects on structured and unstructured data with the goal of building machine learning algorithms capable of generating automated insights for clients.

- Processed large volumes of unstructured text data with regular expressions for cleaning and information extraction, significantly increasing data availability for decision-making.
- Implemented rule-based outlier detection algorithms to flag operational anomalies, contributing to risk mitigation and more reliable strategic decisions.
- Developed a predictive model to estimate the probability of delivery delays, using ranking-based metrics to prioritize the highest-risk orders and enable preventive actions in logistics operations.

## SELECTED PROJECTS

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- **Transaction Fraud Detection** — Classification model to identify fraudulent transactions on highly imbalanced data, with feature engineering, feature selection using Boruta and hyperparameter tuning. The final XGBoost model achieved 91.5% balanced accuracy and 94.4% precision on unseen data. Project with over 220 stars on GitHub. ([github.com/juniorcl/transaction-fraud-detection](https://github.com/juniorcl/transaction-fraud-detection))
- **E-commerce Delivery Time Forecast (Olist)** — LightGBM regression model to predict order delivery time, with data consumed via SQL from a Docker-containerized database, feature selection with mRMR, optimization with Optuna and interpretability with SHAP. Achieved a 66% MAE reduction over the existing business rule, deployed as a Flask API. ([github.com/juniorcl/olist-delivery-forecast](https://github.com/juniorcl/olist-delivery-forecast))
- **DSToolkit — Python Data Science Library** — Modular library with a scikit-learn-inspired API, bringing together AutoML pipelines for classification, regression and clustering, evaluation metrics (ROC, PR, KS, calibration curves, lift-based metrics), residual diagnostics and feature engineering utilities. ([github.com/juniorcl/data-science-toolkit](https://github.com/juniorcl/data-science-toolkit))
- **Rossmann Store Sales Forecast** — Regression model to forecast daily store sales up to six weeks in advance, including exploratory analysis and the formulation and validation of business hypotheses.
- **Health Insurance Cross-Sell** — Customer ranking model to increase the efficiency of insurance cross-selling campaigns, evaluated with cumulative gain and lift curves.

Full portfolio available at [juniorcl.github.io](https://juniorcl.github.io) and [github.com/juniorcl](https://github.com/juniorcl).

## EDUCATION

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### Master's Degree in Natural Sciences — Environmental Science

Universidade Estadual do Norte Fluminense (UNEF) | 2017 – 2019

### Bachelor's Degree in Natural Sciences — Physics

Instituto Federal Fluminense (IFF) | 2010 – 2016

## CERTIFICATIONS AND COURSES

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- Quickstart: LangGraph Essentials — Python
- Clustering: Extracting Patterns from Data
- Text Recognition with OCR and Python
- Data Science Program — Comunidade DS
- Applied Statistics — Prof. Thiago Marques Statistics Community
- Statistics and Optimization for Data Science and Operations Research — USP
- MLOps: Model Deployment and Machine Learning APIs

## LANGUAGES

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- **Portuguese:** Native
- **English:** Full Professional Proficiency
- **Esperanto:** Full Professional Proficiency